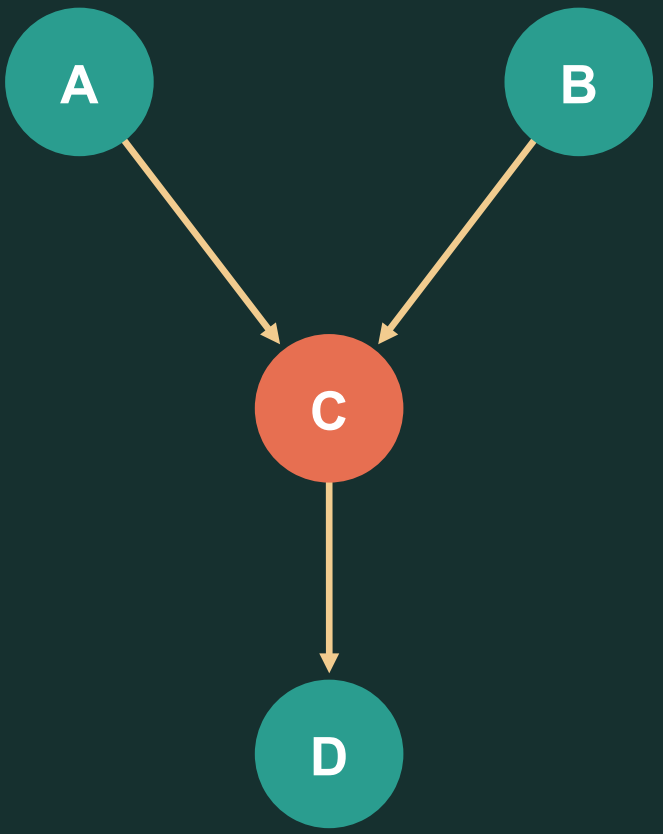
Can a model read causality from text?

Neuro-Symbolic Causal Inference via LLM Knowledge Extraction and Bayesian Networks



DL/AML Project 1 · Team T043 · Section D · 27 October 2026

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The task: is a causal claim entailed?

PREMISE

Suppose there is a closed system of 3 variables, A, B and C. A correlates with C. B correlates with C. However, A is independent of B.

HYPOTHESIS

A directly causes C.

LABEL

Valid. A and B are independent but both correlate with C, so C must be a common effect: A → C ← B in every graph that fits the premise.

6 hypothesis templates → 4 relation types: direct, indirect, collider, confounder

205,626

training rows over 2,336 premises (we train on a fixed 30,000-row sample)

1,162

test rows over 21 premises, 2 to 6 variables each

15.5%

of test hypotheses are valid, so accuracy is misleading; we report F1

One pipeline, three scoring systems

1

Parse

Premise → variables and independence facts. Hypothesis → (relation, X, Y).

Abhijeet

→

2

Score triples

AML baseline, DistilBERT and a symbolic oracle each give P(valid).

Divyanshi · Shubham · Prashant

→

3

Review gate

Keep direct edges above a validation-tuned threshold; drop self loops, two-way pairs and cycles.

Divyanshi

→

4

Build DAGs

One acyclic graph per premise; every edge keeps the row it came from.

Divyanshi

→

5

Bayes net

Fit each graph with pgmpy and answer do-queries against a known gold network.

Rahul

EVALUATED SEPARATELY AT EACH STAGE

Triple extraction
precision, recall, F1 on 1,162 held-out test rows

Graph
edge precision and recall against the gold direct edges

Interventions
|ATE error| on 76 known do-queries per test set

Every stage is run twice: on the official test set and on the same rows with the variable letters shuffled.

Four data problems we found first

5

leaked premises

3 validation and 2 training premises also appear in test. We removed them from validation and train.

0%

positive rate of “B directly causes A”

Causes are always ordered alphabetically, a shortcut. We built a test set with shuffled letters.

37.7%

of training pairs exceed 512 tokens

DistilBERT sees a truncated premise; we cut the premise, never the hypothesis.

44

of 1,162 test labels contradict the oracle

Across 12 premises. We audited them instead of trusting the labels blindly.

Fixing these first is what makes the later comparisons fair: no leakage between splits, a test that exposes the shortcut, and a check on the labels themselves.

Three ways to score a causal triple

A — AML baseline

Pattern features → logistic regression

- 6 features read from the parsed premise: adjacency, marginal correlation, separating sets, common neighbours, size, letter order
- Crossed with the relation type, so one model learns a rule per relation
- Class-balanced; trains in seconds

Divyanshi

D — DistilBERT

Fine-tuned (premise, hypothesis) classifier

- distilbert-base-uncased, 2 epochs, learning rate 5e-5, batch 16
- Same 30,000 training rows as AML
- Max 512 tokens; ~1.5 h on an Apple M3 Pro, resumes from checkpoints

Shubham

O — Symbolic oracle

PC algorithm → equivalence class

- Skeleton from the stated independences, then v-structures
- Enumerate every DAG consistent with the premise
- Valid only if the hypothesis holds in all of them; no training

Prashant

Each system gets one decision threshold, chosen for F1 on validation only. Test is never used for tuning.

Extraction: two text models, a near tie



0.617 vs 0.605

Overall F1, AML vs DistilBERT. The oracle reaches 0.871.

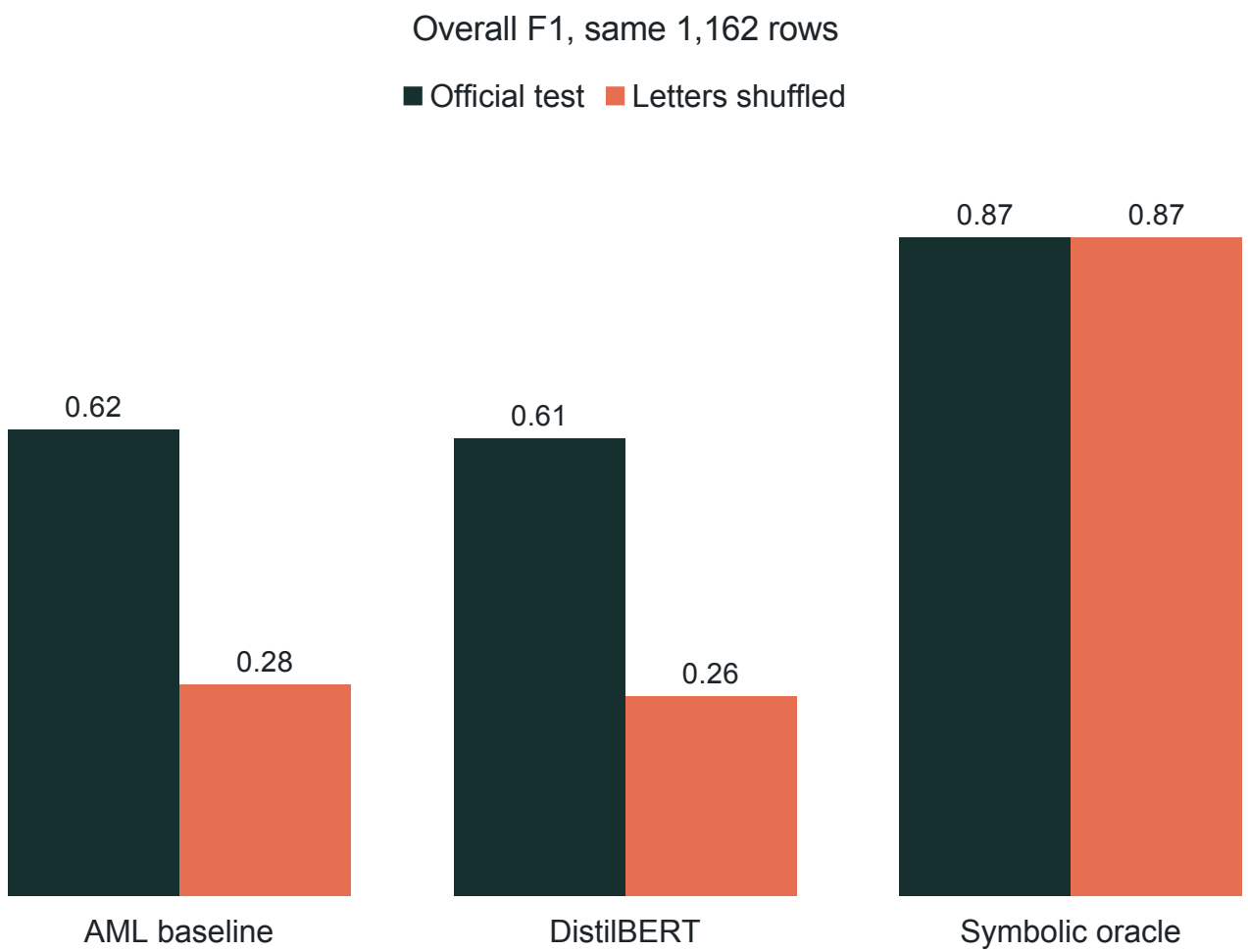
DistilBERT

is far better on confounders (0.743 vs 0.000) and indirect causes (0.657 vs 0.452).

AML

is better on direct edges (0.703 vs 0.610) and colliders (0.667 vs 0.564).

Shuffle the letters and both models fail



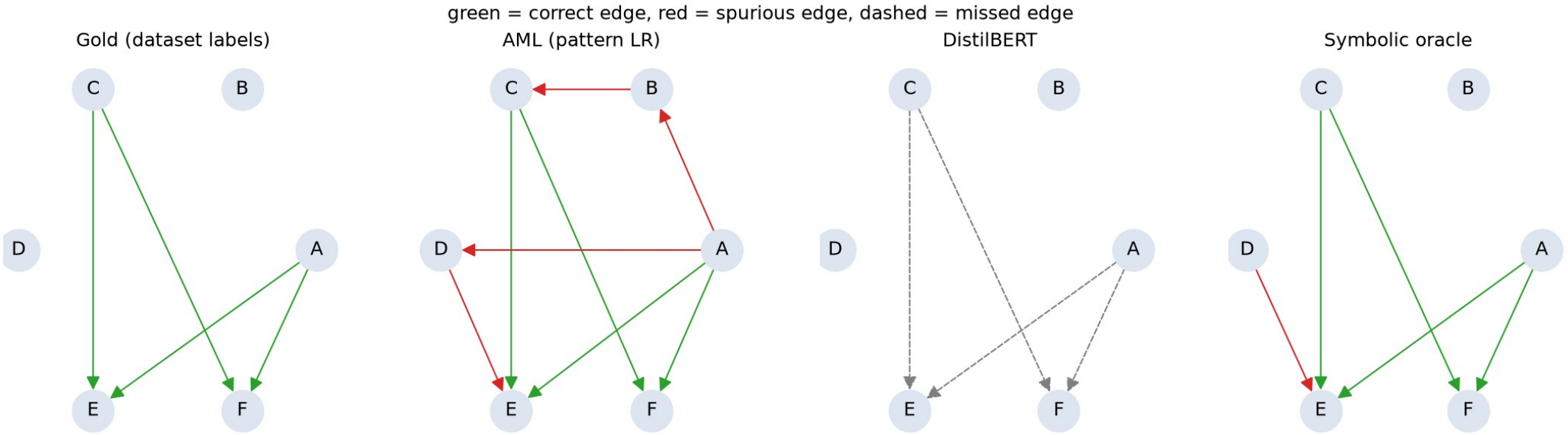
Same premises, same logic; only the variable names are permuted per premise.

AML relies on its letter-order feature, the only one renaming changes.

DistilBERT loses the same cue and also sees premises listed out of alphabetical order, which no training premise does.

The oracle reasons over the graph, so names cannot matter: 0.871 both times.

From triples to one DAG per premise



Test set	Edge precision	Edge recall	Edge F1
AML baseline	0.542	1.000	0.703
DistilBERT	0.505	0.769	0.610
Symbolic oracle	**0.890**	**1.000**	**0.942**

The review rules removed nothing: all 395 edges the text models proposed point from an earlier to a later letter, so they cannot form a cycle. Acyclicity is still enforced and unit-tested.

Bayesian networks and do-queries

1. **Gold network**

   Noisy-OR CPDs on the gold DAG (leak 0.1, strength 0.8)

2. **Simulate**

   5,000 rows from the gold network

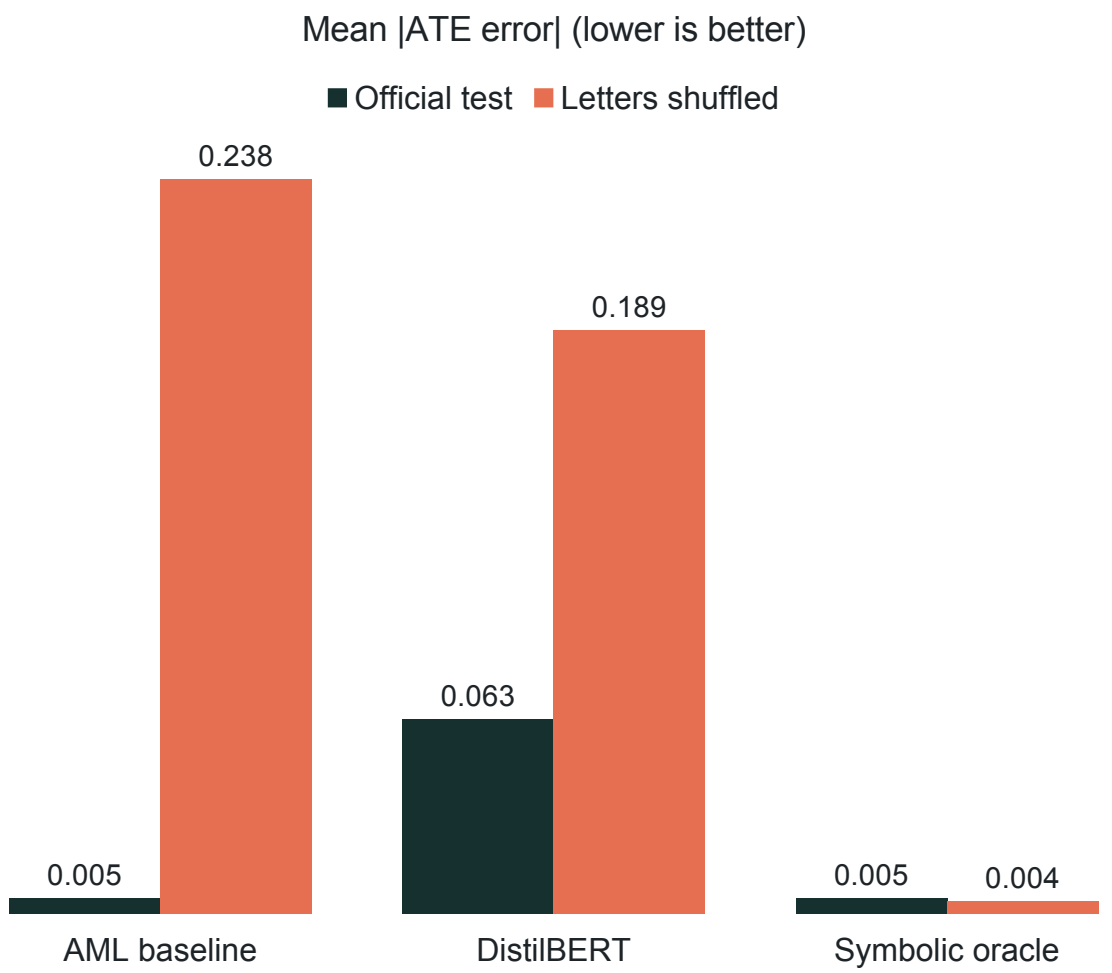
3. **Fit**

   Each system's DAG fitted on the same rows with pgmpy (MLE)

4. **Query**

   ATE = P(Y=1 | do(X=1)) − P(Y=1 | do(X=0)), 76 queries on 5 premises

Extra edges cost little, because the fitted network learns they carry no effect. Missing and reversed edges are what break the queries.



Where the dataset labels are wrong

44

of 1,162 test labels disagree with the symbolic oracle, across 12 of 21 premises

31 labelled valid where the oracle says not entailed; 13 the other way round.

WORKED EXAMPLE

A premise says A and B are independent, also given C or given D, but never given E. So E is a collider, and A → E holds in every DAG consistent with the premise. The dataset still labels “A directly causes E” as invalid.

WHAT THE 10 AUDITED ROWS SHOW

5

labels contradict every DAG consistent with the premise

5

match a looser reading of the relation

99.2%

agreement on indirect causes if read as “a directed path of length ≥ 2”, vs 95.1% for ours

Validation only leaned the same way by one row (333 vs 332 of 337), so we kept our definition fixed rather than tune it on test.

What the gap looks like

1

Text models learn the surface, not the logic

Both reach F1 ≈ 0.61 and both fall to ≈ 0.27 when only the variable names change.

2

Structure carries the reasoning

The oracle keeps edge F1 0.942 and near-zero ATE error whatever the names. The graph stage enforces acyclicity; the BN answers do-queries consistently.

3

Downstream, missing edges hurt most

Spurious edges barely move the ATE; missed and reversed edges do. DistilBERT's misses cost it 0.063 vs 0.005.

Limitations

- Label noise: 44 test labels disputed
- 37.7% of training pairs truncated at 512 tokens
- 30,000 of 205,626 training rows, one seed
- Synthetic CPDs: ATE error tests structure, not the world

Next steps

- Re-sorted renamed test to separate the two shortcut effects
- Longer-context encoder for 6-variable premises
- PubMed abstracts as an open-text extension

Team and contributions

A

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Project setup and codebase structure, data layer (splits, parsers, letter shuffling), end-to-end notebook and results, presentation and report

PP

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Symbolic oracle (PC algorithm, Markov equivalence class, entailment) and the label audit

DK

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AML baseline, graph stage (review gate, DAG per premise, edge metrics) and the edge review

SJ

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DistilBERT fine-tuning and inference, resumable training

RK

Rahul Kumar

2401020100

Bayesian networks and intervention queries (noisy-OR gold network, pgmpy fitting, ATE)

Code: c2c/ package, 25 unit tests, one reproducible notebook (project1_causal.ipynb).

Thank you

Questions?

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